

Matrix functions and stability

Degree coverage with seven matrix multiplications

Difficulty: challenging **Importance:** interesting to the community**Status:** Partially resolved

Rating rationale: Challenging because numerical evidence for coefficient-space coverage needs exact algebraic justification; community impact comes from multiplication budgets for practical matrix-function evaluation.

Verified partial result — 2026-09-11

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A fixed seven-product scheme has a full-rank complex coefficient map, certified by a nonzero exact integer Jacobian minor. Its image contains a nonempty Zariski-open subset of $\mathbb{C}[x]_{\leq 42}$ and is Euclidean dense there.

The upper bound excluding degree 43 and above is not proved. The maximal-degree equality remains open; neither exact representation of every polynomial nor real Euclidean dense coverage is asserted. The entry remains in the open count. Its difficulty and importance ratings continue to describe the surviving canonical target.

Primary reference: [complete authored PDF](#), [standalone TeX](#), [Theorem 1 and equations \(1\)-\(2\)](#). [Independent proof review](#) · [Authorship and submission record](#). Verification is independent agent review, not external human peer review or formal certification.

Problem statement

Start with the scalar polynomials $1, x \in \mathbb{C}[x]$. Allow arbitrarily many complex linear combinations of already computed polynomials and at most seven multiplications of two already computed polynomials. Let $\mathcal{P}_7 \subseteq \mathbb{C}[x]_{\leq 128}$ be the set of possible outputs.

Identify a polynomial with its coefficient vector in $\mathbb{C}^{129}$ and let $\overline{\mathcal{P}_7}^Z$ be its Zariski closure: the common zero set of all polynomial equations in the coefficients that vanish throughout $\mathcal{P}_7$. Is

$$\max\{d \in \{0, \dots, 128\} : \mathbb{C}[x]_{\leq d} \subseteq \overline{\mathcal{P}_7}^Z\} = 42?$$

The target concerns closure and the complex coefficient field. Exact representability of every polynomial is a stronger assertion.

Why it matters

For a dense input matrix, matrix products dominate the cost of polynomial evaluation. The conjecture determines how much polynomial degree a budget of seven products can cover, providing a theoretical benchmark for matrix-function evaluation schemes.

References

1. E. Jarlebring and G. Lorentzon, *The polynomial set associated with a fixed number of matrix-matrix multiplications*, arXiv:2504.01500v3 (13 August 2025), §2.2 (closure convention), §5.4, Conjecture 13. [Primary text](#).
2. J. Sastre, J. Ibáñez, J. M. Alonso, and E. Defez, *Beyond Paterson–Stockmeyer: Advancing matrix polynomial computation*, WSEAS Transactions on Mathematics 24 (2025), 684–693, §4 (a five-product construction). [Primary paper](#), [DOI](#).
3. J. M. Alonso, J. Sastre, J. Ibáñez, and E. Defez, *A systematic framework for stable and cost-efficient matrix polynomial evaluation*, arXiv:2603.23143v1 (24 March 2026), abstract and conclusions. [Primary text](#).

Status check — 2026-09-08

The latest arXiv source remains v3 and labels the displayed equality a conjecture supported by computations. The later papers give constructions and stability procedures without establishing the seven-product degree-coverage claim. Searches included the exact title with 2026, *Jarlebring Lorentzon conjecture seven multiplications* 42, and *Sastre polynomial evaluation* 2026. No resolution was located. A [3 September 2026 author talk](#) also describes higher-cost degree coverage as conjectural. Six-product real/complex formulations have a separate source ambiguity and are deliberately not included here.

Audit — 2026-09-10

Rechecked [Conjecture 13](#) and [Remark 14](#): complex closure coverage at seven products is still conjectural, with concrete matrix-exponential applications. Title and seven-product searches found no exact proof. The [June 2026 degree-eight construction](#) solves a different budget/degree problem.